The PHP Company

Collaborative Open Source with PHP on IBM i

Mike Pavlak

Solution Consultant

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Agenda

- **What's up with MySQL?**
- **Mediawiki**
  - Installation
  - Configuration
  - Operation
- **Concrete5**
  - Installation
  - Configuration
  - Operation
- **Q&A**

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Collaborative Open Source with PHP for IBM i

www.zend.com

What‘s up with MySQL?

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MySQL

- MySQL discontinued distributions last year.
- Zend and IBM are partnering to bring something back!
- Get ready for…

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Zend DBi

- Zend Technologies will provide Zend DBi as a drop-in replacement for MySQL on the IBM i platform
- Zend DBi will be shipped as a part of Zend Server for IBM i and freely downloadable as a standalone package
- The build will remain open-source and be updated over time to implement relevant bug fixes and features
- IBM will continue to maintain and deliver the DB2 Storage engine for MySQL for IBM i, via Zend Server and as a standalone package

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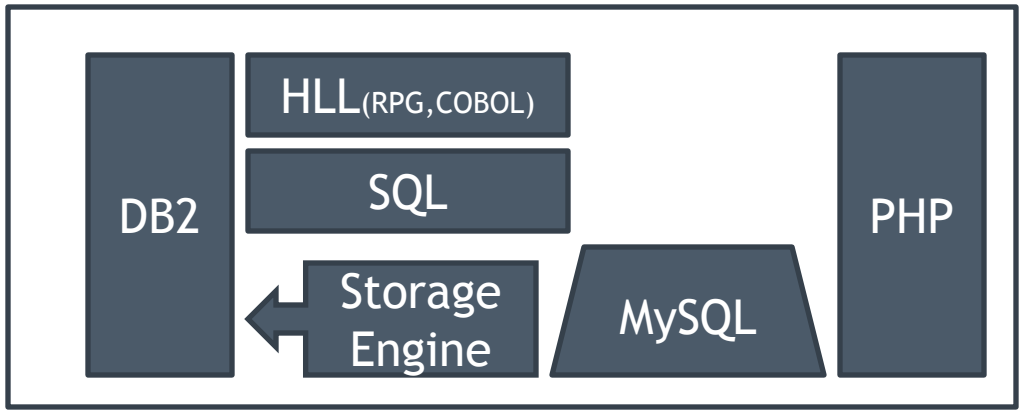
Why Zend Dbi?

- COMMON, COMMON Europe and WMCPA (Wisconsin Midrange Computer Professional Association) all use open source application Joomla
- Manage their respective organizations with PHP handling the transactions and presentation layer.
- These organizations depend on MySQL as the database storing content and important transaction processing.



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Collaborative Open Source with PHP for IBM i

www.zend.com

Media Wiki

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zend

Mediawiki

Key Features

- Great for user annotated content
- Documentation, anyone?

- **Installation on IBM i**
  - i7.1
  - Current group/cume PTF's
  - Zend Server 5.1 running PHP 5.3.6
- **Business Application:**
  - Easy to maintain user docyumentation
  - Users can add comments right to the page!!!

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MediaWiki

MediaWiki

www.mediawiki.org/wiki/MediaWiki

Log in / create account

Page | Discussion | Read | View source | View history | Search

MediaWiki

Main page
Browse categories
Community portal
Recent changes
Tech blog

Support
User help
Technical manual
FAQ
Support desk

Download
Development
Communication
Print/export
Toolbox

Welcome to MediaWiki.org

MediaWiki is a free software open source wiki package written in PHP, originally for use on Wikipedia. It is now used by several other projects of the non-profit Wikimedia Foundation and by many other wikis, including this website, the home of MediaWiki.

Use the links below to explore the site contents. You'll find some content translated into other languages, but the primary documentation language is English.

For general questions about MediaWiki see the communication page or ask at the support desk.

About this site | **About MediaWiki** | **Download** | **Help and support**

Users

- **What is a wiki?**
- Learn **how to navigate**
- Learn **how to edit a page**
- Get **more help**

System administrators

- **Install** & **configure** MediaWiki
- **Upgrade** an existing MediaWiki installation
- Add features with **third-party extensions**
- Get **more sysadmin help**

Developers

- **Become a MediaWiki hacker**
- Learn to use the **API** and **write extensions**
- Browse the **developer docs** and **class reference**
- Already a hacker? Visit the **developer hub**

Current version

- **1.17.0** – 2011-06-22

News

- **2011-06-22: 1.17.0 released**

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Get the file

- Navigate to the downloads page
- Pull down the latest copy
- I used 7-zip to uncompress the .gz
- FTP (FileZilla) the .tar to the root file system of the IFS
- Untar using QP2TERM
  - tar –xvf mediawiki-1.17.0.tar
  - This unpacks all the files to a directory of the same name
- Point your browser at the index.php in the app directory

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Session A - [24 x 80]

File Edit View Communication Actions Window Help

Host: zend.idevcloud.com Port: 23 Workstation ID: mikep11 Disconnect

```
                    /QOpenSys/usr/bin/-sh

  x mediawiki-1.17.0/maintenance/sqlite/archives/patch-rename-iwl_prefix.sql, 1
  85 bytes, 1 media blocks.
  x mediawiki-1.17.0/maintenance/sqlite/archives/searchindex-no-fts.sql, 757 by
  tes, 2 media blocks.
  x mediawiki-1.17.0/maintenance/README, 3066 bytes, 6 media blocks.
  x mediawiki-1.17.0/php5.php5, 174 bytes, 1 media blocks.
  x mediawiki-1.17.0/mw-config
  x mediawiki-1.17.0/mw-config/index.php5, 64 bytes, 1 media blocks.
  x mediawiki-1.17.0/mw-config/index.php, 1229 bytes, 3 media blocks.
  x mediawiki-1.17.0/trackback.php, 1785 bytes, 4 media blocks.
  x mediawiki-1.17.0/profileinfo.php, 8893 bytes, 18 media blocks.
  x mediawiki-1.17.0/load.php5, 27 bytes, 1 media blocks.
  x mediawiki-1.17.0/README, 4057 bytes, 8 media blocks.
  $

===> tar -xvf mediawiki-1.17.0.tar

F3=Exit     F6=Print    F9=Retrieve    F11=Truncate/Wrap
F13=Clear   F17=Top     F18=Bottom     F21=CL command entry

MA    A                MW                                    18/007
```

I902 - Session successfully started

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Installation…MediWiki.org

Manual:Installation guide - Me

www.mediawiki.org/wiki/Manual:Installation_guide

Log in / create account

Manual | Discussion | Read | Edit | View history | Search

MediaWiki

Main page
Browse categories
Community portal
Recent changes
Tech blog

Support
User help
Technical manual
FAQ
Support desk

Download

Development

Communication

Print/export

Toolbox

Manual:Installation guide

Installation Guide | About MediaWiki | Features | Installation requirements | Downloading | Installing | Configuring

Welcome to the **MediaWiki installation guide** which provides instructions on how to manually install and configure MediaWiki on a web server. Note that some users may find MediaWiki software bundles a convenient alternative to manual installation.

The first part of this guide gives broad general instructions, while the last section contains a list of appendices that give more detailed installation notes for specific system configurations and other less common uses of the software.

Contents [hide]

1 Quick installation guide
2 Main installation guide
3 Appendices
3.1 Advanced uses
3.2 Advanced configuration
3.3 MediaWiki Help
3.4 System-specific instructions
4 See also

Quick installation guide [edit]

For experienced users, here is the quick version of the installation instructions. Most users will want to read the more detailed version, below.

1. Check that your system meets the minimum requirements (web server, PHP5 (**do not use 5.3.1**; 5.3.2+ is OK, 5.3.3 is released),

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Not setup yet, click the link…

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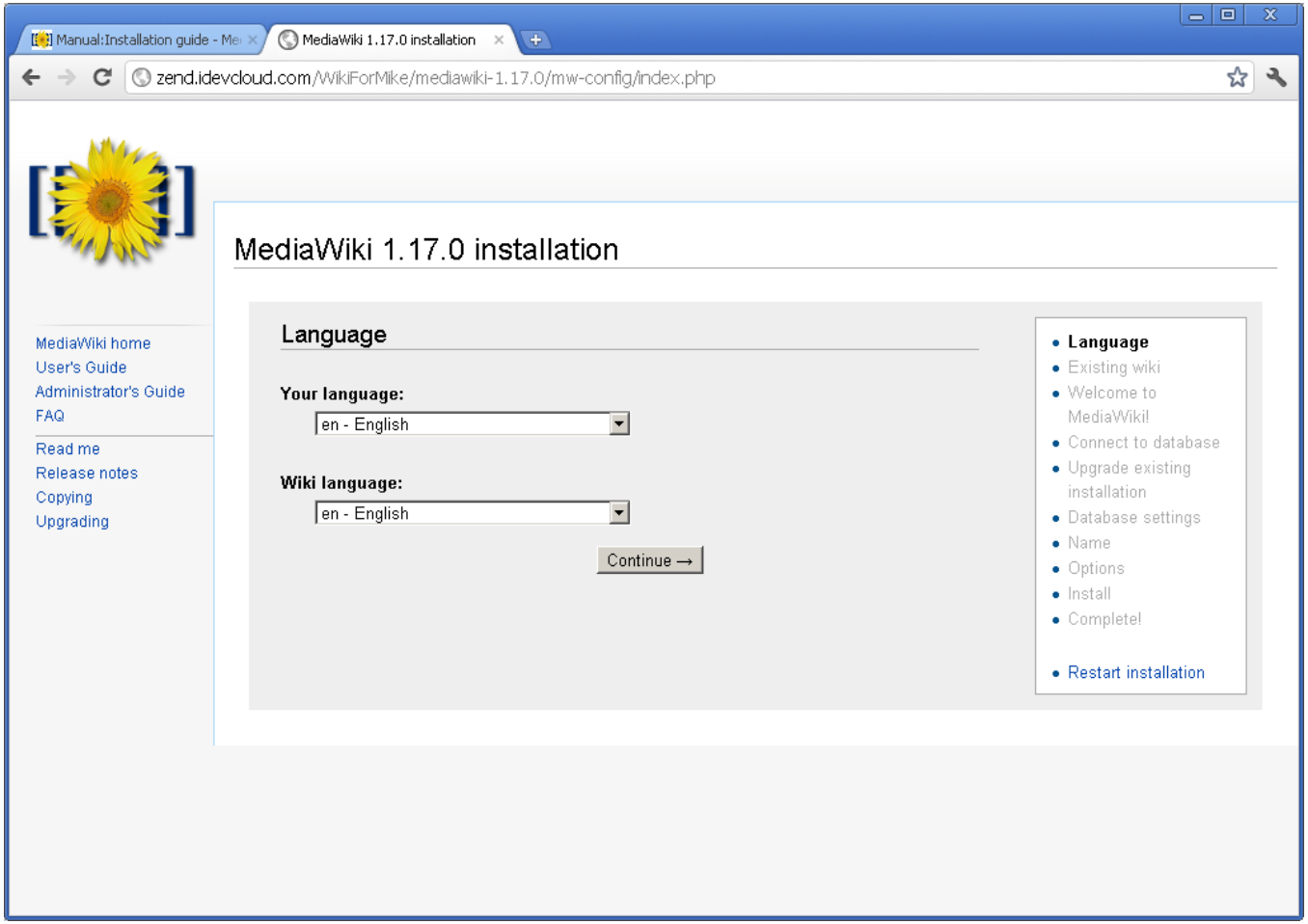
Default to English



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Environment is OK

Manual:Installation guide - Me
MediaWiki 1.17.0 installation

zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=Welcome

MediaWiki home
User's Guide
Administrator's Guide
FAQ

Read me
Release notes
Copying
Upgrading

MediaWiki 1.17.0 installation

Welcome to MediaWiki!

- Language
- Existing wiki
- **Welcome to MediaWiki!**
- Connect to database
- Upgrade existing installation
- Database settings
- Name
- Options
- Install
- Complete!

- Restart installation

Environmental checks

Basic checks are performed to see if this environment is suitable for MediaWiki installation. You should provide the results of these checks if you need help during installation.

- PHP 5.3.6 is installed.
- GNU diff3 not found.
- Found GD graphics library built-in.

Image thumbnailing will be enabled if you enable uploads.

- **Warning:** Your default directory for uploads `/www/zendsvr/htdocs/WikiForMike/mediawiki-1.17.0/images/` is vulnerable to arbitrary scripts execution.

Although MediaWiki checks all uploaded files for security threats, it is highly recommended to close this security vulnerability before enabling uploads.

- Using the intl PECL extension for Unicode normalization.
- **Warning**: The installed version of the Unicode normalization wrapper uses an older version of the ICU project's library.

You should upgrade if you are at all concerned about using Unicode.

The environment has been checked. You can install MediaWiki.

Copyright and Terms

This wiki is powered by **MediaWiki**, copyright © 2001-2011 Magnus Manske, Brion Vibber, Lee Daniel Crocker, Tim Starling, Erik Möller, Gabriel Wicke, Ævar Arnfjörð Bjarmason, Niklas Laxström, Domas Mituzas, Rob Church, Yuri Astrakhan, Aryeh Gregor, Aaron Schulz, Andrew Garrett, Raimond Spekking, Alexandre Emsenhuber, Siebrand Mazeland, Chad Horohoe, Roan

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Setup the database…

Manual:Installation guide - Me
MediaWiki 1.17.0 installation

zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=DBConnect

MySQL settings

Database host:
help
localhost

Identify this wiki

Database name:
help
wikiformike

Database table prefix:
help
wm

User account for installation

Database username:
help
mpavlak

Database password:
help
••••••

← Back | Continue →

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Database settings…

Manual:Installation guide - Me
MediaWiki 1.17.0 installation
zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=DBSettings

MediaWiki 1.17.0 installation

MediaWiki home
User's Guide
Administrator's Guide
FAQ
Read me
Release notes
Copying
Upgrading

Database settings

Database account for web access

Use the same account as for installation

Database username:
wikiformike

Database password:
••••••••

help

Create the account if it does not already exist

Storage engine:
InnoDB
MyISAM

help

Database character set:
Binary
UTF-8

help

← Back Continue →

- Language
- Existing wiki
- Welcome to MediaWiki!
- Connect to database
- Upgrade existing installation
- **Database settings**
- Name
- Options
- Install
- Complete!
- Restart installation

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Name…

Manual:Installation guide - Me
MediaWiki 1.17.0 installation
zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=Name

Copying
Upgrading

Project namespace:
help
- Same as the wiki name: Wikiformike
- Project
- Other (specify)

Administrator account

Your name:
help
mpavlak

Password:
•••••••

Password again:
•••••••

E-mail address:
help
mike.p@zend.com

help
Subscribe to the release announcements mailing list.

You are almost done! You can now skip the remaining configuration and install the wiki right now.

installation
- Database settings
- **Name**
- Options
- Install
- Complete!

- Restart installation

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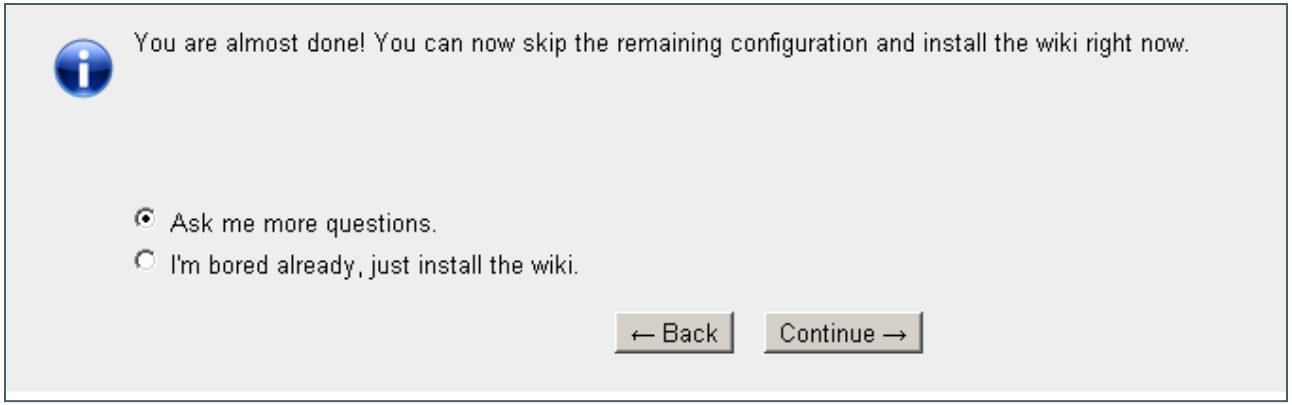
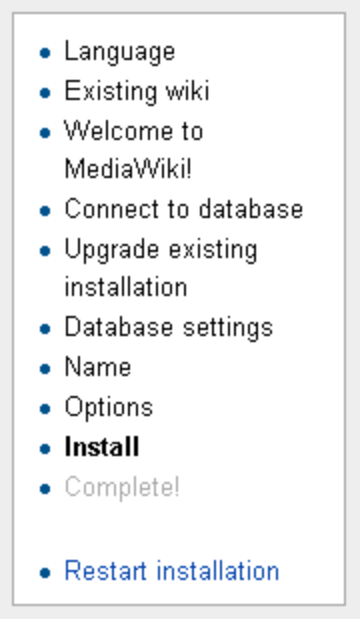
Are you bored yet>

- The heavy lifting is done…
- Click radio button as you see fit…





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Results, not bad…

Manual:Installation guide - Me
MediaWiki 1.17.0 installation
zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=Install

MediaWiki 1.17.0 installation

MediaWiki home
User's Guide
Administrator's Guide
FAQ
Read me
Release notes
Copying
Upgrading

Install

- Setting up database... done
- Creating tables... done
- Creating database user... done
- Populating default interwiki table... done
- Initializing statistics... done
- Generating secret keys... done
- Creating administrator user account... done

Unable to subscribe to mediawiki-announce: Error fetching URL: SSL certificate problem, verify that the CA cert is OK. Details: error:14090086:SSL routines:SSL3_GET_SERVER_CERTIFICATE:certificate verify failed

- Creating main page with default content... done

Continue →

- Language
- Existing wiki
- Welcome to MediaWiki!
- Connect to database
- Upgrade existing installation
- Database settings
- Name
- Options
- **Install**
- Complete!

- Restart installation

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Download the local settings file…

Manual:Installation guide - Me
MediaWiki 1.17.0 installation
zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/mw-config/index.php?page=Complete

MediaWiki home
User's Guide
Administrator's Guide
FAQ
Read me
Release notes
Copying
Upgrading

MediaWiki 1.17.0 installation

Complete!

Congratulations! You have successfully installed MediaWiki.

The installer has generated a `LocalSettings.php` file. It contains all your configuration.

You will need to download it and put it in the base of your wiki installation (the same directory as index.php). The download should have started automatically.

If the download was not offered, or if you cancelled it, you can restart the download by clicking the link below:

Download LocalSettings.php

Note: If you do not do this now, this generated configuration file will not be available to you later if you exit the installation without downloading it.

When that has been done, you can **enter your wiki**.

- Language
- Existing wiki
- Welcome to MediaWiki!
- Connect to database
- Upgrade existing installation
- Database settings
- Name
- Options
- Install
- **Complete!**

- Restart installation

LocalSettings (1).php
LocalSettings.php
Show all downloads…

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Local settings, a look inside

PHP - C:\Users\mike.p\Downloads\LocalSettings.php - Zend Studio - C:\Users\mike.p\Zend\workspaces\DefaultWorkspace7

File Edit Source Refactor Navigate Search Project Run Window Help

ToolkitService.php | calc2.php | calc1.php | PrecoatCall.php | PrecoatSprintf.php | PrecoatSprintf.php | orderReport.php | LocalSettings.php

```php
<?php
# This file was automatically generated by the MediaWiki 1.17.0
# installer. If you make manual changes, please keep track in case you
# need to recreate them later.
#
# See includes/DefaultSettings.php for all configurable settings
# and their default values, but don't forget to make changes in _this_
# file, not there.
#
# Further documentation for configuration settings may be found at:
# http://www.mediawiki.org/wiki/Manual:Configuration_settings

# Protect against web entry
if ( !defined( 'MEDIAWIKI' ) ) {
    exit;
}

## Uncomment this to disable output compression
# $wgDisableOutputCompression = true;

$wgSitename      = "wikiformike";
$wgMetaNamespace = "Wikiformike";

## The URL base path to the directory containing the wiki;
## defaults for all runtime URL paths are based off of this.
## For more information on customizing the URLs please see:
## http://www.mediawiki.org/wiki/Manual:Short_URL
$wgScriptPath       = "/WikiForMike/mediawiki-1.17.0";
$wgScriptExtension  = ".php";

## The relative URL path to the skins directory
$wgStylePath        = "$wgScriptPath/skins";

## The relative URL path to the logo.  Make sure you change this from the default,
## or else you'll overwrite your logo when you upgrade!
$wgLogo             = "$wgStylePath/common/images/wiki.png";
```

Writable | Smart Insert | 1 : 1

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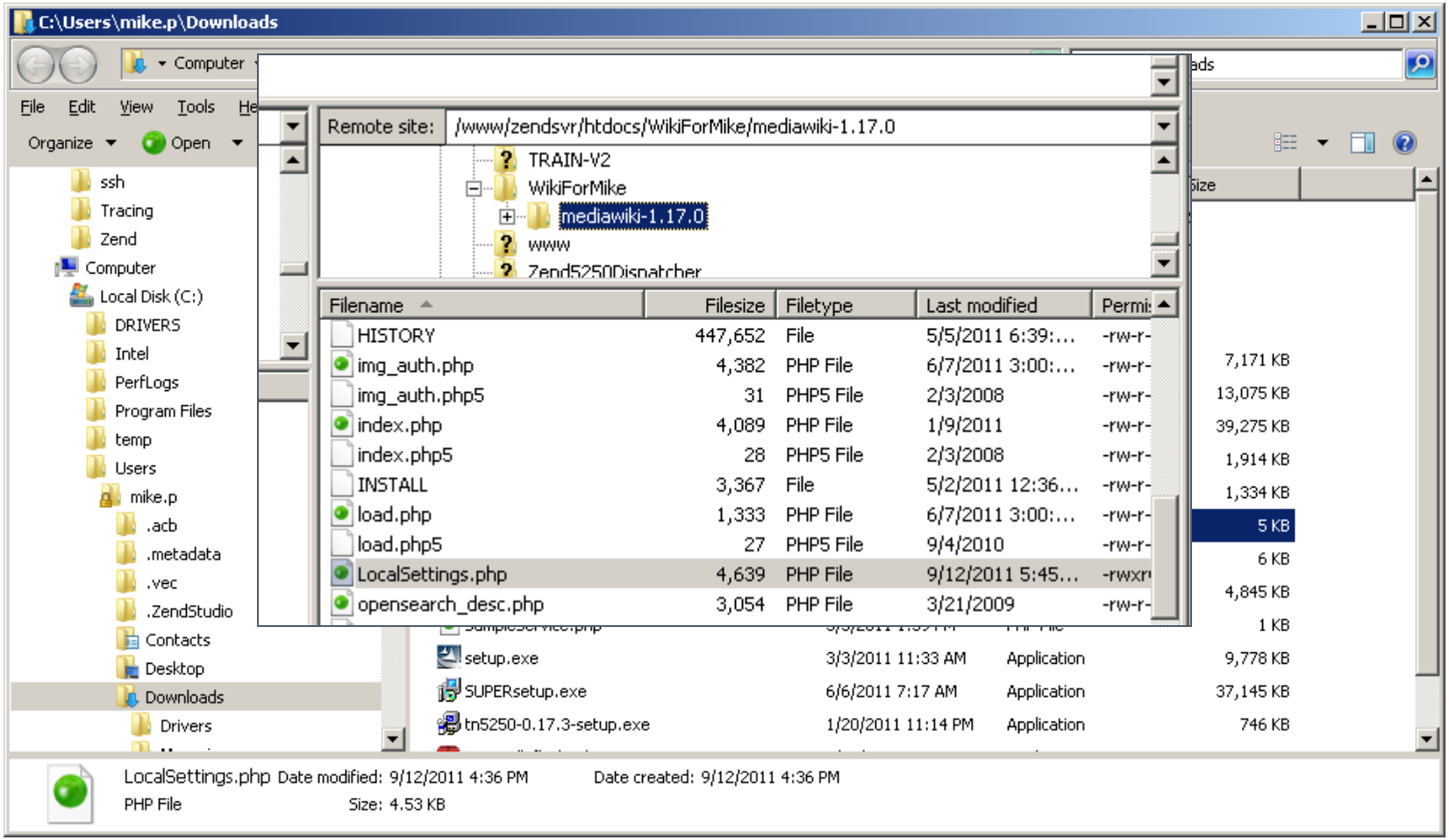
Used File-Zilla to copy files…



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Tada!

Manual:Installation guide - Me
wikiformike
zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/index.php?title=Main_Page

Log in / create account

Set $wgLogo to the URL path to your own logo image.

Page Discussion Read Edit View history Go Search

Main Page

MediaWiki has been successfully installed.

Consult the User's Guide for information on using the wiki software.

Getting started [edit]

- Configuration settings list
- MediaWiki FAQ
- MediaWiki release mailing list

Navigation

Main page
Community portal
Current events
Recent changes
Random page
Help

Toolbox

What links here
Related changes
Special pages
Printable version
Permanent link

This page was last modified on 12 September 2011, at 21:33.

Privacy policy About wikiformike Disclaimers

Powered By MediaWiki

LocalSettings (1).php LocalSettings.php Show all downloads...

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Housekeeping…

- Logo?
  - Change the $wgLogo value in LocalSettings.php
  - Edit with Zend Studio
  - Copy file to local directory in MediaWiki
    - WikiForMike/mediawiki-1.17.0/skins/common/images

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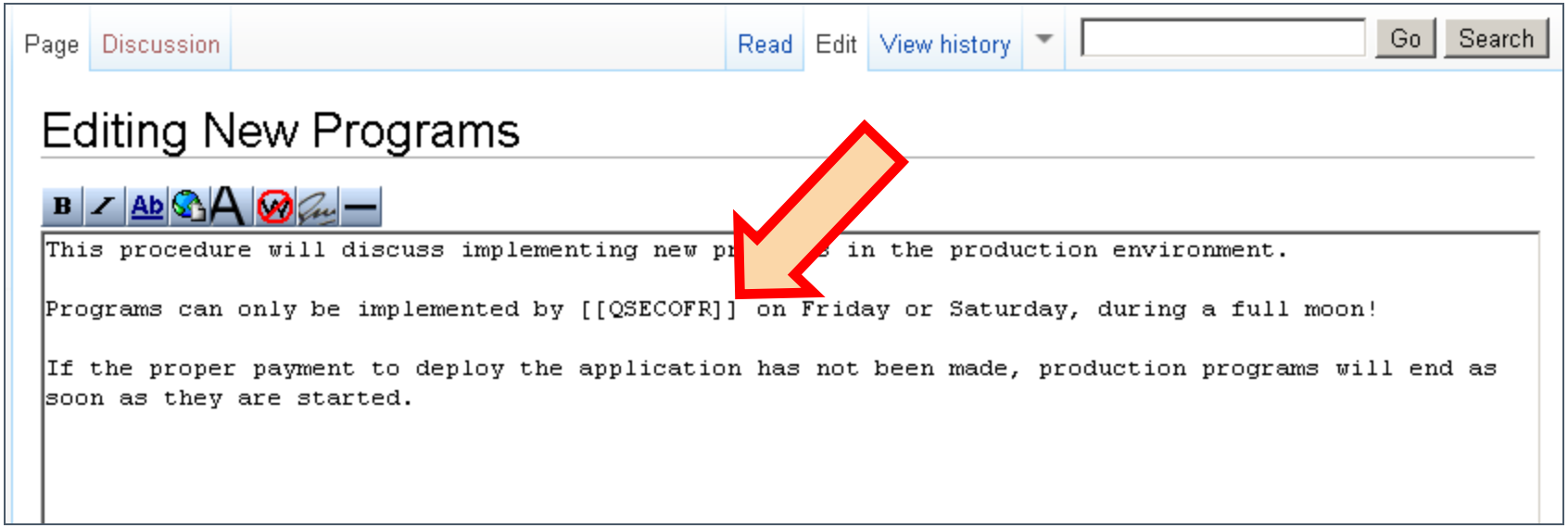
New page topic…

- **Search for it**
  - MediaWiki will let you create if it does not exist
  - Can control creation upon login
  - Once created, can link to others



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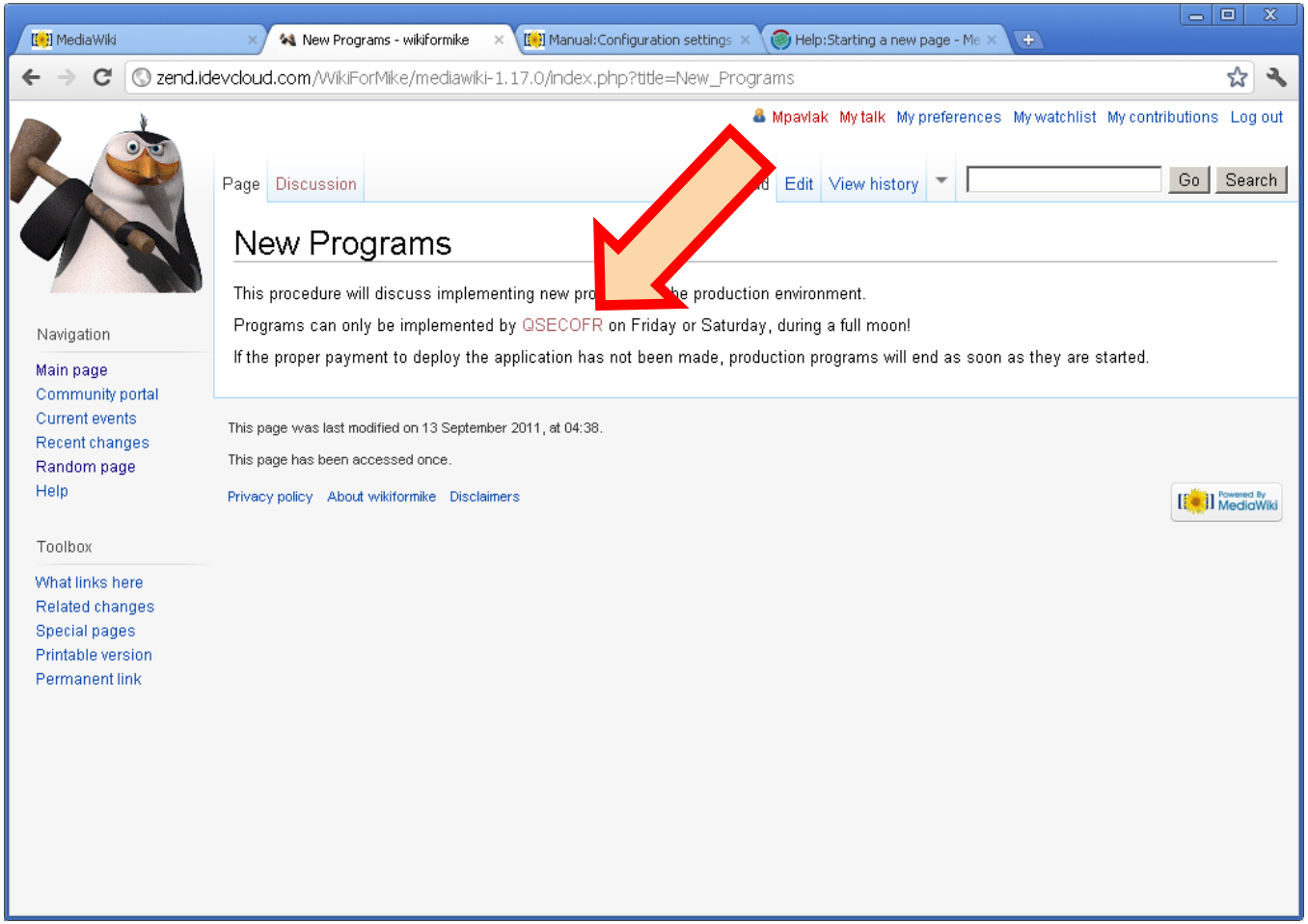
Finished page looks like…



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Linked Page

- A page to edit QSECOFR comes up
- Enter contents and save

Page Discussion Create Go Search

Editing QSECOFR

You have followed a link to a page that does not exist yet. To create the page, start typing in the box below (see the help page for more info). If you are here by mistake, click your browser's **back** button.

QSECOFR is the most powerful user profile in the IBM i world. Once you let it out it is VERY hard to get it back!

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New QSECOFR Page

MediaWiki | QSECOFR - wikiformike | Manual:Configuration settings | Help:Starting a new page - Me

zend.idevcloud.com/WikiForMike/mediawiki-1.17.0/index.php?title=QSECOFR

Mpavlak My talk My preferences My watchlist My contributions Log out

Page Discussion Read Edit View history Go Search

QSECOFR

QSECOFR is the most powerful user profile in the IBM i world. Once you let it out it is VERY hard to get it back!

Navigation

Main page
Community portal
Current events
Recent changes
Random page
Help

Toolbox

What links here
Related changes
Special pages
Printable version
Permanent link

This page was last modified on 13 September 2011, at 04:40.

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Discussion…

- As a developer you created QSECOFR page.
- Let users annotate the contents with the discussion tab

Page Discussion Read Edit Add topic View history Go Search

Talk:QSECOFR

yeah, it may be powerful, but the password in under the keyboard in the computer room!

Password changed! [edit]

We changed the password and hid it somewhere else, HA!

This page was last modified on 13 September 2011, at 04:43.

This page has been accessed once.

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Collaborative Open Source with PHP for IBM i

www.zend.com

Concrete5

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Concrete5

Key Features

- PHP, Zend Framework, jQuery
- Documentation, anyone?

- **Installation on IBM i**
  - i7.1
  - Current group/cume PTF's
  - Zend Server 5.1 running PHP 5.3.6
- **Business Application:**
  - Intranet, Internet content site
  - Department messaging

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Why Concrete5

- **Companies standardizing on Zend Framework**
- **Also standardizing on JavaScript libraries**
- **Clean separation between…**
  - Editors (Content)
  - Programmers (Code & function)
  - Designers (Look and feel)
- **Quick and easy startup**

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Home page

concrete5 - Free CMS | Open :
MikeBio
concrete5 :: Unable to connec

www.concrete5.org

Join Now | Sign In | Cart | Instant Setup | Download

Search all of concrete5.org

concrete5

About | Community | Developers | Marketplace | Services | Documentation

CMS made for Marketing but built for Geeks

concrete5 is a content management system that is free and open source.

Sitemap
Full Sitemap
File Manager
Sitemap
Reports

Site Editors

When your website is built with concrete5, you easily change content from anywhere in the world.

Learn More

Developers

concrete5 is a free open source content management system written in PHP.

Learn More

Designers

Templating concrete5 is a breeze. With a few lines of code your designs will be editable in minutes.

Learn More

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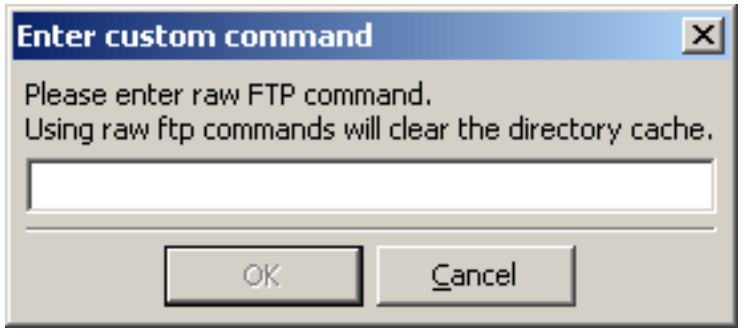
Installation…

- Copy files from download up to the IBM i
- Unzip locally
- FileZilla made quick work of upload
  - TIP: If in FileZilla and you cannot get directory listing try…
    - Server→Enter Custom Command…
    - Put in SITE NAMEFMT 1
    - Refresh server output panel (right click)



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Installation…

- Create an empty MySQL database
- Create a MySQL User Profile
- Open browser & point to directory with Concrete5
- Installation instructions:
  - Page: http://www.concrete5.org/documentation/installation/installing_concrete5/
  - Forum: http://www.concrete5.org/community/forums/installation/
  - YouTube videos for Conrete5 embedded in site: http://www.concrete5.org/

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Installation

concrete5 :: Basic Setup | zend.idevcloud.com/concrete5

zend.idevcloud.com/concrete5/conapp/con5421/

Install concrete5

Testing Required Items

- PHP 5.1 Available
- JavaScript Enabled
- MySQL Available
- Support for C5 Request URLs
- Image Manipulation Available
- XML Support
- Web Server Access to Files and Configuration Directories

Testing Optional Items

- Remote File Importing Available
- Version Comparison Available

Run Tests Again

Having trouble? Check the installation help forums, or have us host a copy for you.

Personal Information

Name Your Site: ConZend

Your Email Address: mike.p@zend.com

Database Information

Server: localhost

MySQL Username: concrete5user

MySQL Password:

Database Name: concreteuser

Sample Content

☑ Install sample content

Install concrete5 >

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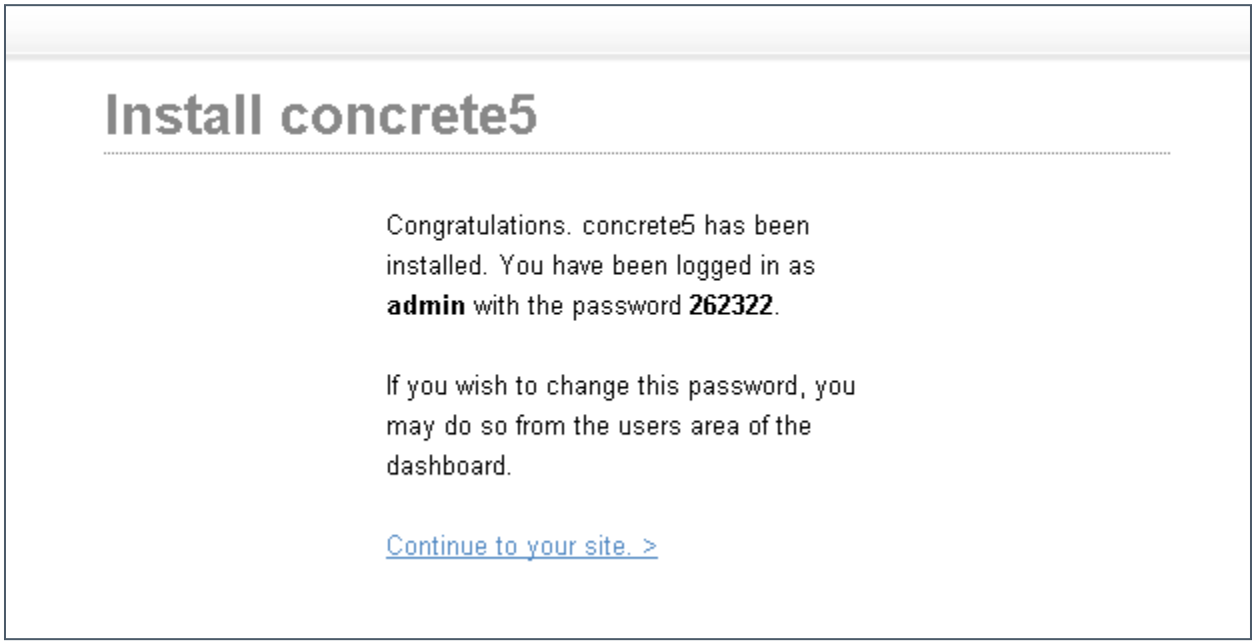
Installation successful



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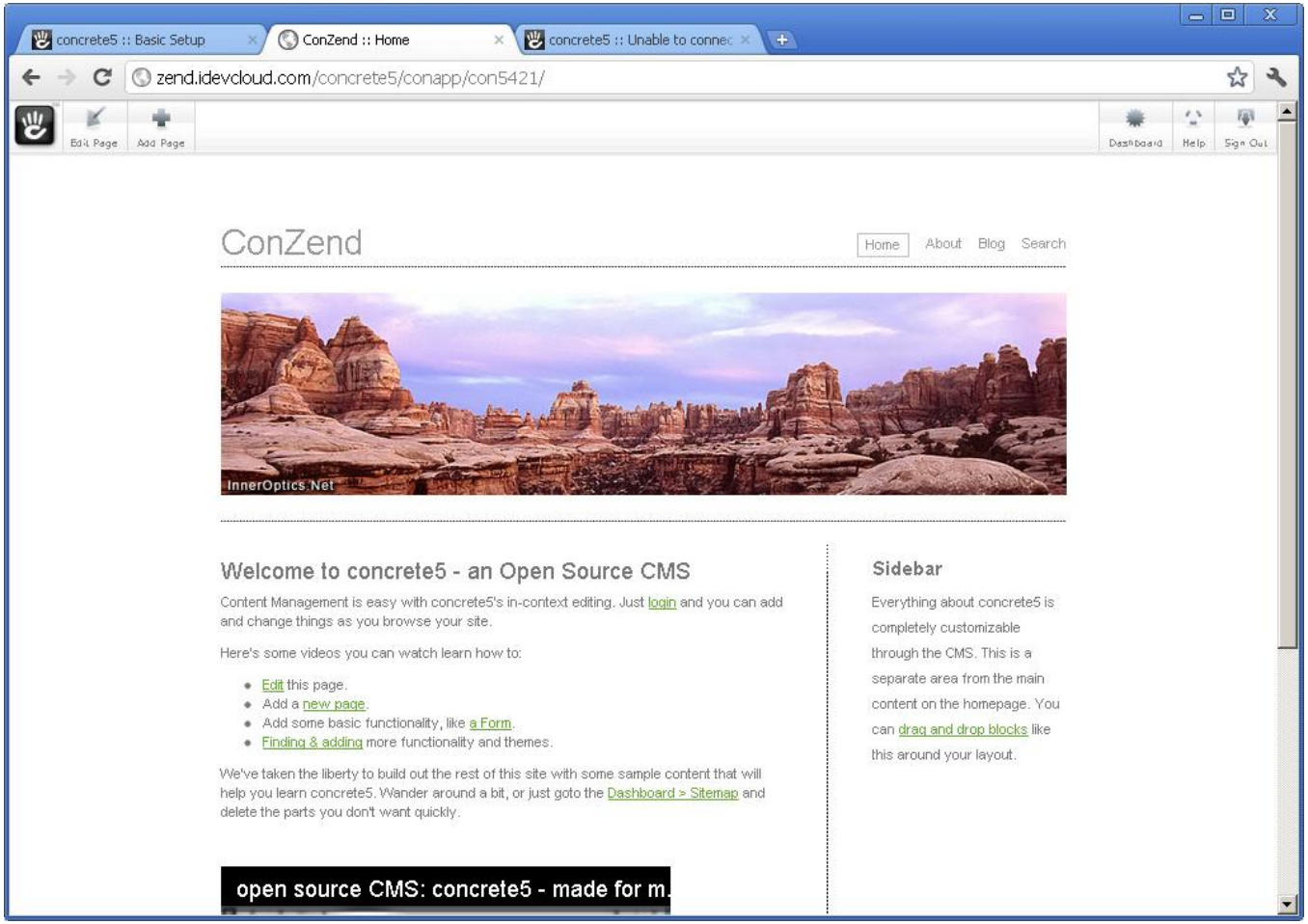
Main page…



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Dashboard

concrete5 :: Basic Setup | ConZend :: Dashboard | concrete5 :: Unable to connec

zend.idevcloud.com/concrete5/conapp/con5421/index.php/dashboard/

Return to Website | Help | Sign Out

Version: 5.4.2.1

Composer Beta
Write for your site.

Sitemap
Whole world at a glance.

File Manager
All documents and images.

Reports
Get data from forms and logs.

Users and Groups
Add and manage people.

Scrapbook
Share content across your site.

Pages and Themes
Reskin your site.

Add Functionality
Install addons & themes.

System & Maintenance
Backup, cleanup and update.

Sitewide Settings
Secure and setup your site.

Site Activity

You are logged in as admin. You logged in on 09/13/11 at 5:44 AM.

Number of visits since your previous login: 0

Total visits: 0

Total page versions: 84

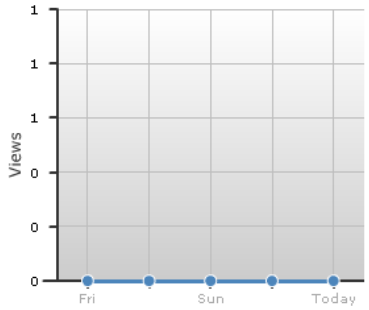
Last edit: 09/13/11 at 1:44 AM

Last login: Your login is the most recent.

Total pages in edit mode: 0

Total form submissions: 0 today (0 total)

Statistics



Help

Search Documentation

Search

Full Documentation

Full documentation is available at Concrete5.org.

Latest News

View all >

There are no notifications.

Notes

Write notes to yourself using the text area below.

Save

zend.idevcloud.com/concrete5/conapp/con5421/index.php/tools/r...

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Advantages

- Added a page
- Edit content
- You Tube videos for education
- In 5 minutes I was able to…
  - Load the site
  - Adjust the security
  - Walked the configuration
  - Edited and deployed my bio

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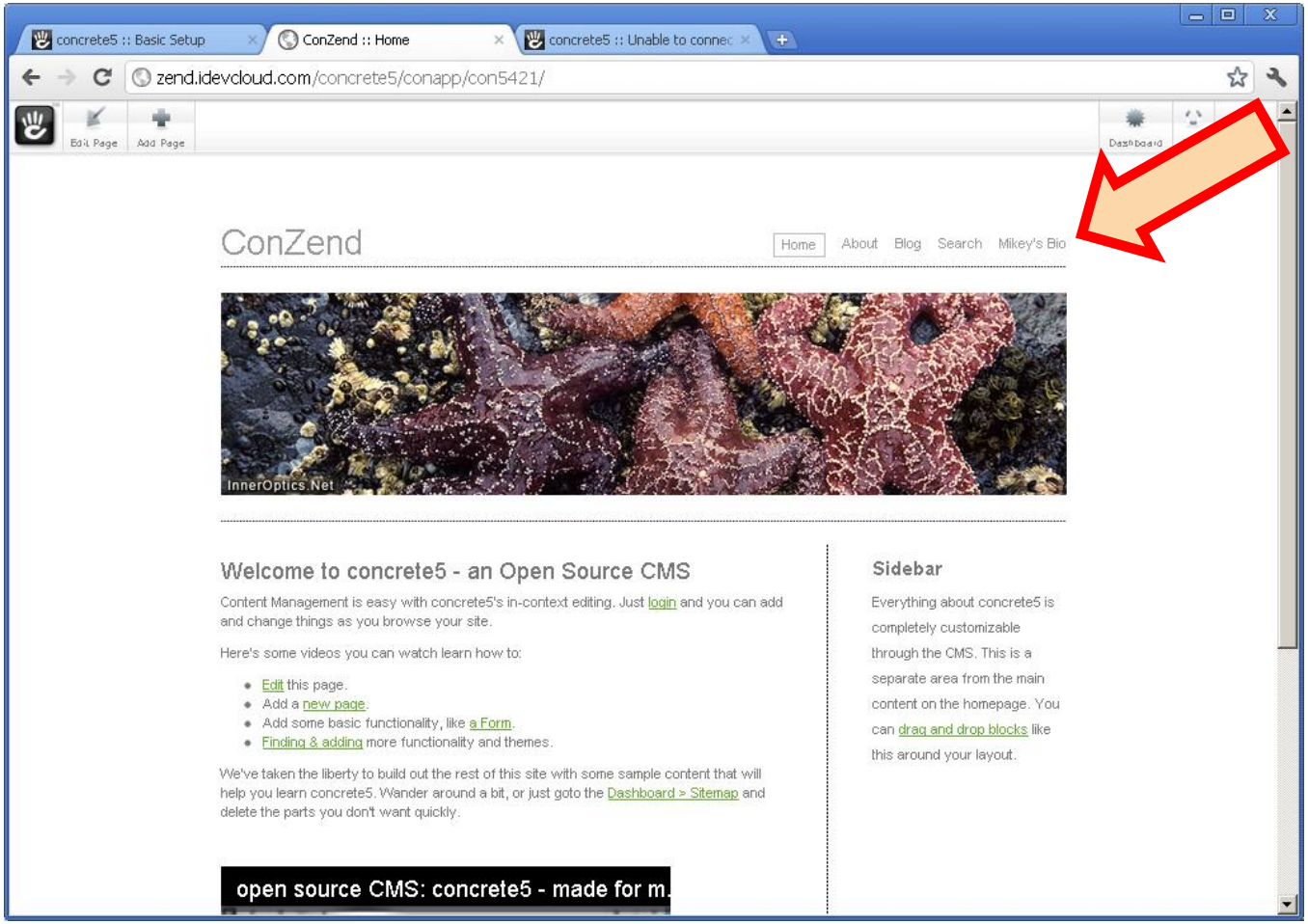
Home page



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Mikey's Bio

concrete5 :: Basic Setup | MikeBio | concrete5 :: Unable to connec

zend.idevcloud.com/concrete5/conapp/con5421/index.php/mikey-s-bio/

Edit Page | Add Page | Dashboard | Help | Sign Out

ConZend

Home About Blog Search Mikey's Bio

Mikey's Bio for Public Speaking

Mike Pavlak has been working with IBM midrange solutions since 1992 and IBM Mainframes before that. After years of developing applications using RPG, CL and PHP he managed IT development and IT for power protection manufacturer Tripp Lite. Mike joined Zend Technologies as a Solutions Consultant working with the IBM sales team in 2008 and enjoys helping IBM i customers explore PHP as a state of the art solution for application modernization. In addition to several roles as a volunteer with COMMON the premier IBM midrange user group, Mike also teaches PHP classes part time at Moraine Valley Community College in Suburban Chicago.

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concrete5 - open source CMS

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Collaborative Open Source with PHP for IBM i

www.zend.com

Wrap-up

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Upcoming Webinars

- **September**
  - Open Source on IBM i Part 2
- **October**
  - Best Practices for deploying secure PHP applications
- **More soon…**

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Conference Themes

Cloud Computing
Learn about the latest developments in PHP Cloud infrastructure, management and application services

Mobile and User Experience
Learn how to build engaging mobile apps with the latest PHP technologies and tools

Enterprise and Professional PHP
Explore PHP best practices, new technologies and practical tips with industry experts

Conference Highlights

- Sessions focused on how to best develop and deploy PHP
- Sessions designed for all knowledge levels
- Intensive tutorials for accelerated learning
- PHP Certification crash courses and testing
- Exhibit hall showcasing the latest products
- Special networking opportunities during meals and events

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Reasons to attend ZendCon

- **IBM i track with AWESOME features like**
  - PHP Toolkit authors
  - Sessions on performance, etc.
- **Lots of PHP that applies to IBM i shops, too**
  - http://mikepavlak.blogspot.com/2011/08/zendcon-for-mere-mortals.html
- **IBM i Evening Event**
  - Special networking opportunity for PHP developers devoted to IBM i
- **Fabulous Keynotes**

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Q&A

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